

Exoplanetary Rover SLAM

Problem ID: problem

A fleet of autonomous rovers is exploring a subterranean cave system on exoplanet Kepler-16b. Each rover runs a SLAM (Simultaneous Localisation and Mapping) algorithm tracking N distinct geological landmarks, maintained in an $N \times N$ state transition matrix.

During the latest exploration phase, each rover recorded M sequential state transitions $M_1, M_2, \dots, M_M$, capturing how the map evolved step by step. The navigation module, which is prone to memory corruption from the cave's intense magnetic interference, has produced a matrix E representing its current belief of the map state after applying all transitions in sequence. The rovers use E to plan their next move and if it is wrong, they are navigating on a corrupted map and risk driving straight into a cave wall.

Before giving the rovers the green light, you must verify from ground control that E truly equals $M_1 \times M_2 \times \dots \times M_M$. Every second the rovers sit idle drains their battery, so you need an answer fast.

Transmitting full matrices is too slow, so each rover instead transmits only the seed used to generate each transition matrix. You reconstruct them locally using a shared PRNG: given seed S , the entries are filled in row-major order with rows and columns indexed from 0 using:

$$M_{ij} = (S \cdot 42 + i \cdot 7 + j \cdot 3) \bmod 100$$

Input

The first line contains an integer $1 \leq N \leq 400$, the matrix dimension. The second line contains an integer $1 \leq M \leq 60$, the number of transition matrices. The following M lines each contain a single integer $1 \leq S_i \leq 10^9$, the seed for matrix M_i . The final N lines each contain N space-separated integers describing the current map state matrix E .

Output

Output YES if the rovers can safely advance, otherwise output NO.

Sample Input 1	Sample Output 1
2 2 5 12 183 252 288 399	YES

Sample Input 2	Sample Output 2
2 2 686579304 119540832 6613 7062 7278 7737	NO